Physiology Practice Block: Neural and Hormonal Regulation of the GIT
Multiple-Choice Questions (40 MCQs)

Q1. The Enteric Nervous System (ENS) is a highly complex, autonomous neural network contained entirely within the gastrointestinal wall. What structural characteristic allows the ENS to coordinate gut motility and secretion completely independent of the central nervous system?
a. It consists entirely of unmyelinated somatic afferent nerve arcs.
b. It contains its own complete reflex pathways including sensory receptors, interneurons, and motor neurons embedded directly inside the tissue layers.
c. It functions via a direct non-synaptic connection to the thoracic skeletal motor pools.
d. It relies on the conversion of electrical tracking waves into volatile gas fractions.
e. It acts exclusively under the direct control of the uncrossed cranial motor roots.

Q2. The myenteric plexus (Auerbach's plexus) is an essential component of the intrinsic enteric network. Where is this specific plexus anatomically situated within the cross-section of the gastrointestinal wall?
a. Between the mucosal epithelium and the muscularis mucosae layer
b. Inside the loose connective tissue stroma of the submucosa
c. Between the longitudinal and circular smooth muscle layers
d. Directly beneath the outer single-cell epithelial layer of the serosa
e. Surrounding the vascular endothelial walls of the mesenteric arterioles

Q3. While the myenteric plexus dominates motor commands, the submucosal plexus (Meissner's plexus) controls localized tissue demands. What is the primary physiological function integrated by the submucosal plexus?
a. Controlling downstream propulsion of food during high-amplitude mass movements
b. Regulating local intestinal secretion, absorption, and localized blood flow along the mucosal folds
c. Maintaining the high tonic baseline tone of the external anal sphincter
d. Adjusting the systemic arterial blood volume via pressure natriuresis loops
e. Translating structural connective collagen directly into hard bone salts

Q4. Excitatory motor neurons within the myenteric plexus are responsible for driving smooth muscle contraction to propagate peristaltic waves. Which fast-acting neurotransmitter substance do these neurons release?
a. Norepinephrine
b. Epinephrine
c. Gamma-aminobutyric acid (GABA)
d. Glycine
e. Acetylcholine

Q5. Coordinated intestinal motility requires that smooth muscle sphincters relax ahead of an advancing food bolus. Which specialized gas and neuro-peptide are released by enteric inhibitory motor neurons to trigger this relaxation?
a. Oxygen gas and somatomedin C fragments
b. Hydrochloric acid vapors and pro-insulin loops
c. Nitric oxide (NO) and Vasoactive Intestinal Peptide (VIP)
d. Carbon dioxide gas and carbamino proteins
e. Nitrogen dioxide and alpha-crystallin molecules

Q6. Sensory nerve fibers line the gastrointestinal tract to sample the internal state of the gut. Which of the following sets of parameters lists the primary stimuli that excite these intrinsic sensory receptors?

a. Distension of the gut wall, mechanical irritation of the mucosa, or specific chemical parameters like acidity and osmolarity in the lumen

b. Exposure of the outer skin surface to ambient ultraviolet radiation wavelengths

c. Absolute, unvarying reduction in the core temperature of distal limbs

d. Shifting of the baseline oxygen-hemoglobin dissociation curve to the left

e. Excessive accumulation of dissolved helium gas fractions in blood plasma

Q7. The baseline electrical activity of gastrointestinal smooth muscle features rhythmic, non-action potential oscillations of the membrane potential. What is the physiological name given to these baseline fluctuations?

a. Spike potentials

b. Slow waves

c. Miniature endplate potentials

d. Threshold drifts

e. Sinusoidal action currents

Q8. Slow waves do not cause muscle contraction directly but dictate the rhythm of gastrointestinal movements. Which specialized pacemaker cells are responsible for generating these slow waves within the muscular layers?

a. Enterochromaffin-like (ECL) cells

b. Chief cells of the mucosal stroma

c. Parafollicular C cells

d. Interstitial Cells of Cajal (ICC)

e. Granular alveolar pneumocytes

Q9. What structural features connect the Interstitial Cells of Cajal to neighboring smooth muscle cells, allowing the pacemaker slow waves to rapidly propagate through the muscle syncytium?

a. Heavy structural desmosome anchors with zero electrical conductance

b. Gap junctions, which provide low-resistance pathways for ionic current flow

c. Active transport pumping lines that exchange sodium for copper coordinates

d. Crystalline hydroxyapatite bridges anchored to the cytoskeleton

e. Specialized synaptic junctions utilizing vesicular norepinephrine

Q10. True mechanical contractions of the gut smooth muscle take place only when spike potentials are superimposed on the peaks of slow waves. What specific ion channel mediates the depolarization during these spike potentials?

a. Fast voltage-gated sodium channels identical to skeletal muscle axons

b. Ligand-gated chloride channels that drive immediate hyperpolarization

c. Slow calcium-sodium channels, which allow large amounts of calcium ions to enter the cell

d. Mechanically gated magnesium channels bound to structural collagen

e. Inward rectifier potassium channels operating in a vacuum

Q11. Autonomic regulation of the gut relies on a balance between extrinsic inputs. How does parasympathetic stimulation generally modulate gastrointestinal activity?

a. It induces complete flaccid paralysis of the entire digestive tract.

b. It shuts down all mucosal blood flow via intense alpha-1 vasoconstriction.

c. It increases the overall activity of the enteric nervous system, augmenting motility and secretion.

d. It decreases the frequency of slow waves to zero inside the pylorus.

e. It acts exclusively on the skeletal muscles controlling the bony orbit.

Q12. The cranial parasympathetic division supplies extensive innervation to the upper gastrointestinal tract. Which cranial nerve carries these preganglionic fibers to the stomach, pancreas, and small intestine?

a. Facial nerve (Cranial Nerve VII)
b. Glossopharyngeal nerve (Cranial Nerve IX)
c. Vagus nerve (Cranial Nerve X)
d. Hypoglossal nerve (Cranial Nerve XII)
e. Trigeminal mandibular branch (Cranial Nerve V3)

Q13. The distal half of the large intestine, including the sigmoid colon and rectum, receives its parasympathetic supply from a different anatomical region. Which nerves carry these signals to coordinate defecation reflexes?

a. Splanchnic nerves arising from the celiac ganglion
b. Pelvic nerves originating from the sacral spinal cord segments (S2, S3, S4)
c. Phrenic nerve motor somatic axon branches
d. Greater petrosal nerve autonomic pathways
e. Intercostal neural networks running along ribs

Q14. How does sympathetic stimulation generally affect gastrointestinal smooth muscle motility and mucosal secretion?

a. It triggers an explosive, immediate surge in all peristaltic waves.
b. It expands the total cross-sectional dead space of the stomach five-fold.
c. It transforms the mucosal epithelium into a stratified squamous shield.
d. It generally inhibits gastrointestinal activity, causing relaxation of smooth muscle and constriction of sphincters.
e. It deactivates the breakdown of liver glycogen at rest.

Q15. Extrinsic postganglionic sympathetic nerve fibers release norepinephrine into the gut wall. Through what primary mechanism does this release inhibit gastrointestinal motility?

a. By a direct minor inhibitory effect on the smooth muscle itself, and a major inhibitory effect on the neurons of the ENS to block acetylcholine release
b. By exciting the pacemaking interstitial cells of Cajal to fire at 100 Hz
c. By forcing all local blood vessels to double their internal diameters
d. By digesting the local myenteric plexus matrix with lysosomal enzymes
e. By opening large, ligand-gated chloride channels on enterocytes

Q16. Gastrointestinal reflexes can be classified into three distinct categories based on their neural pathways. Which of the following is a classic example of a reflex that is integrated entirely within the gut wall enteric nervous system?

a. The gastrocolic reflex driving colonic mass movements
b. The enterogastric reflex blocking stomach emptying
c. Local reflexes that coordinate peristalsis, mixing contractions, and localized secretion in a single segment of the bowel
d. The pain reflex causing generalized inhibition of the entire GI tract
e. The defecation reflex traveling from the rectum to the spinal cord

Q17. The second category of GI reflexes travels from the gut to the prevertebral sympathetic ganglia and then projects back to other regions of the tract. What is the name of the reflex where distension of the stomach drives rapid evacuation of the colon?

a. The enterogastric reflex

b. The gastrocolic reflex
c. The colono-gastric reflex
d. The gastroileal reflex
e. The rectospinal reflex

Q18. Which reflex travels from the duodenum back to the stomach to inhibit gastric motility and secretion when the duodenal loop is filled with highly acidic chyme?

a. The gastrocolic reflex
b. The duodenocolic reflex
c. The enterogastric reflex
d. The colono-gastric reflex
e. The gastroileal reflex

Q19. The third category of GI reflexes travels from the gut to the spinal cord or brainstem and then returns to the gastrointestinal tract. What is a classic example of this long, central reflex loop?

a. Local segmenting contractions within the jejunum loop
b. The vagovagal reflex, where sensory signals from the stomach travel to the brainstem and return via the vagus nerve to control gastric motility and secretion
c. Secretion of mucus by a single isolated goblet cell
d. Passive movement of water across the tight junctions
e. The intrinsic myenteric defecation reflex arc

Q20. When an individual experiences severe pain, inflammation, or trauma within the abdominal cavity, a protective reflex shuts down almost all gut motility. What is this reflex called?

a. The gastrocolic reflex
b. The intestinal-intestinal inhibitory reflex (or paralytic ileus reflex) operating via sympathetic centers
c. The vagovagal secretory surge loop
d. The myogenic autoregulatory bounce
e. The enterogastric feedback current

Q21. Gastrointestinal smooth muscle displays a continuous, low-level contraction that lasts for minutes or hours, independent of the rhythmic slow waves. What type of contraction pattern is this?

a. Rhythmic segmented twitching
b. Saccadic jumping sweep
c. Tonic contraction
d. Graded electrotonic drift
e. Asynchronous explosive discharge

Q22. The resting membrane potential of gastrointestinal smooth muscle is highly variable. Which of the following occurrences shifts the membrane potential in a positive direction, depolarizing it and making the tissue more excitable?

a. Release of epinephrine by the adrenal medulla acting on beta receptors
b. Stimulation of sympathetic postganglionic fibers releasing norepinephrine
c. Stretching of the muscle wall, or the release of acetylcholine from parasympathetic nerve endings
d. Excessive accumulation of nitrogen gas bubbles inside capillaries
e. Active clearance of all available intracellular calcium by transport pumps

Q23. What specific electrical change is induced in the gastrointestinal smooth muscle membrane when it is exposed to circulating epinephrine or intense sympathetic nerve discharge?

a. Immediate, high-velocity depolarization triggering spike bursts
b. Hyperpolarization, which shifts the membrane potential in a negative direction and reduces muscle excitability
c. Intracellular locking of the voltage baseline at positive 30 millivolts
d. Complete structural dissolution of all active gap junctions
e. Fluctuations that exactly duplicate the mechanical heart rate frequency

Q24. While slow waves establish the maximum frequency of contractions, they do not cause muscle contraction on their own. Why is a slow wave unable to trigger mechanical contraction in the absence of spike potentials?

a. Because slow waves do not cause calcium ions to enter the smooth muscle fiber; they only cause a slow influx of sodium ions
b. Because slow waves completely deactivate all intracellular calmodulin proteins
c. Because slow waves drive the membrane potential to a highly negative baseline
d. Because they operate exclusively inside non-contractile fibrous sheets
e. Because slow waves are restricted to the mucosal epithelium layer

Q25. What primary intracellular molecular mechanism links an increase in free calcium ions during a spike potential to actual mechanical contraction inside a gut smooth muscle cell?

a. Calcium binds to troponin C, which shifts tropomyosin away from actin filaments
b. Calcium binds to calmodulin, and this complex activates myosin light chain kinase (MLCK) to phosphorylate the myosin cross-bridges
c. Calcium activates proteolytic enzymes that physically shorten the cytoskeleton
d. Calcium opens ligand-gated chloride tracks to drive structural shortening
e. Calcium transforms the thin microfilaments into hard compact bone salts

Q26. Gastrointestinal hormones are synthesized by enteroendocrine cells to regulate gut function. Which hormone is released by G cells in the stomach antrum primarily to stimulate gastric acid secretion and mucosal growth?

a. Cholecystokinin (CCK)
b. Secretin
c. Gastrin
d. Glucose-Dependent Insulinotropic Peptide (GIP)
e. Motilin

Q27. When fatty acids, monoglycerides, and products of protein digestion enter the duodenum, they stimulate I cells to release a hormone that contracts the gallbladder and stimulates pancreatic enzyme secretion. What is this hormone?

a. Gastrin
b. Secretin
c. Motilin
d. Cholecystokinin (CCK)
e. Somatostatin

Q28. Highly acidic chyme (pH less than 4.5 to 5.0) entering the duodenal loop stimulates S cells to release a hormone into the blood. What is this hormone, and what is its primary target tissue action?

a. Gastrin; stimulating acid production in the stomach

b. Secretin; stimulating pancreatic duct cells to secrete a large volume of fluid rich in bicarbonate ions to neutralize acid
c. CCK; stimulating gallbladder contraction and bile flow
d. Motilin; driving the migrating motor complexes during fasting
e. GIP; triggering immediate insulin release from the islets

Q29. Glucose-Dependent Insulinotropic Peptide (GIP) belongs to the enterogastrone class of hormones. Which cells secrete GIP, and what is its dual metabolic and digestive role?

a. I cells; contracting the gallbladder and blocking pancreatic fluid flow
b. S cells; releasing bicarbonate and neutralizing gastric secretions
c. G cells; driving acid production and stimulating antral motility
d. K cells; stimulating insulin secretion from beta cells in response to oral glucose, and mildly inhibiting gastric motility and acid secretion
e. D cells; universally suppressing all endocrine and exocrine gut secretions

Q30. Which gastrointestinal hormone is secreted by specialized cells in the duodenal and jejunal mucosa during fasting states to drive the cyclical cleansing waves of the migrating motor complex?

a. Gastrin
b. Cholecystokinin
c. Secretin
d. GIP
e. Motilin

Q31. The Enteric Nervous System can continue to coordinate the complex motor pattern of peristalsis even if all extrinsic sympathetic and parasympathetic nerves are surgically severed. What rule of motility describes this phenomenon?

a. The myogenic autoregulatory bounce
b. The law of the gut (or the peristaltic reflex), where a contractile ring forms on the oral side of a bolus while the anal side relaxes
c. The Hering-Breuer defensive lung inflation loop
d. The gastrocolic mass movement synchronization rule
e. The stress-relaxation pressure accommodation law

Q32. While the ENS operates with a high degree of autonomy, a complete congenital absence of enteric ganglion cells can occur in a segment of the large intestine. What clinical condition develops as a result?

a. Achalasia of the cardia smooth muscle
b. Congenital hypertrophic pyloric stenosis
c. Hirschsprung's disease (congenital aganglionic megacolon)
d. Barrett's esophagus column metaplasia
e. Zollinger-Ellison gastrinoma syndrome

Q33. Why does an aganglionic segment of the colon cause a severe functional obstruction and proximal megacolon in a child with Hirschsprung's disease?

a. The aganglionic segment cannot undergo reflexive relaxation, creating a permanent, high-resistance barrier that blocks the forward movement of feces.
b. The smooth muscle layer undergoes rapid structural calcification within days.
c. High abdominal pressure blocks the entry of all available liquid volumes.
d. The muscle cells are converted into a loose adipose tissue layer.
e. The segment switches to synthesizing huge amounts of steroid hormones.

Q34. When an individual is placed under extreme, unremitting sympathetic stress or fear (fight-or-flight state), what functional change is observed in their digestive efficiency and mucosal blood flow?

a. Digestion is accelerated to maximize glucose availability to muscles.
b. GI motility and secretion are heavily suppressed, and intense vasoconstriction dramatically reduces splanchnic blood flow to divert blood to vital organs.
c. The gastrointestinal tract walls double their baseline diameters.
d. The smooth muscle layers transform into hard compact bone rings.
e. The enteric nervous system completely digests its own plexus ganglia.

Q35. Blood flow through the gastrointestinal tract is part of a specialized circulatory network. What is the name of the system that collects all venous blood from the gut, pancreas, and spleen and delivers it directly to the liver?

a. The systemic inferior vena cava tract
b. The internal jugular venous drainage line
c. The portal venous system (splanchnic circulation)
d. The thoracic lymphatic duct pathway
e. The cavernous sinus vascular network

Q36. During the digestion of a meal, local gastrointestinal blood flow increases significantly. Which of the following factors coordinates this localized functional hyperemia?

a. Widespread activation of postganglionic sympathetic fibers
b. Severe, progressive depigmentation of the mucosal lining layers
c. Complete structural calcification of the local vascular spurs
d. Immediate structural reduction in the total blood volume across limbs
e. The release of vasodilator hormones (such as CCK, secretin, and gastrin) and local kinins, combined with a drop in oxygen concentration due to increased metabolic activity

Q37. Somatostatin-secreting D cells are located throughout the gastrointestinal mucosa. Through what specific transport pathway does somatostatin reach its target membranes to execute its inhibitory effects?

a. It enters the systemic arterial blood to make a full circulatory loop.
b. It utilizes specialized ATP-driven transport pumps inside long axons.
c. It turns into a volatile gas fraction that diffuses through local bone.
d. It diffuses locally through the interstitial fluid spaces via paracrine signaling.
e. It passes out through the mucosal surface to float in gastric chyme.

Q38. Gastrin and Cholecystokinin share a highly specific structural feature at their carboxyl terminus that is essential for binding their targets. How many amino acids form this identical C-terminal sequence?

a. A sequence of 5 amino acids (Gly-Trp-Met-Asp-Phe-NH2)
b. Exactly two individual protein links
c. Perfectly fixed at forty-five items to match structural indices
d. Over one hundred individual peptide connections
e. None, as their chemical structures are completely unrelated

Q39. What diagnostic test is considered the gold standard to differentiate a autonomous gastrin-secreting tumor (Zollinger-Ellison syndrome) from simple G-cell hyperplasia?

a. Intravenous insulin tolerance challenge
b. The secretin stimulation test (secretin paradoxically increases gastrin in gastrinomas)
c. Complete structural clearance of plasma albumin

d. High-frequency electrical stimulation of the phrenic nerve
e. Voluntary hyperventilation for ten minutes

Q40. Which specific autonomic center located in the medulla oblongata integrates sensory inputs from the gut wall to coordinate long, central vagovagal reflexes?

a. The pneumotaxic center
b. The superior and inferior salivatory nuclei
c. The solitary tract nucleus (nucleus tractus solitarius)
d. The pretectal visual reflex nucleus
e. The lateral vestibular nucleus complex

True/False Statements (15 Statements)

1. The myenteric plexus is located between the circular and longitudinal smooth muscle layers and is primarily responsible for regulating gastrointestinal movements and propulsion.

- **Answer:**

2. The submucosal plexus is situated in the submucosa and its master physiological role is the control of localized intestinal secretion, absorption, and mucosal blood flow.

- **Answer:**

3. Slow waves are true action potentials that cause strong mechanical smooth muscle contractions in the complete absence of any superimposed spike potentials.

- **Answer:**

4. Interstitial Cells of Cajal function as the electrical pacemakers of the gastrointestinal tract, generating rhythmic slow wave oscillations that travel into smooth muscle cells via gap junctions.

- **Answer:**

5. Parasympathetic stimulation to the gut travels via the vagus and pelvic nerves, leading to a generalized increase in the activity of the entire enteric nervous system.

- **Answer:**

6. Postganglionic sympathetic nerve fibers release acetylcholine to directly excite gastrointestinal smooth muscle cells and accelerate peristalsis during exercise.

- **Answer:**

7. The enterogastric reflex is an intrinsic enteric nervous system loop that operates entirely within the stomach wall to speed up gastric emptying when the stomach stretches.

- **Answer:**

8. The gastrocolic reflex is a long reflex arc where distension or irritation of the stomach triggers mass movements in the colon to clear feces after a meal.

- **Answer:**

9. True spike potentials are mediated by the rapid opening of fast voltage-gated sodium channels that open and close within a fraction of a millisecond, identical to skeletal muscle axons.

- **Answer:**

10. Tonic contractions of the gastrointestinal tract are continuous, long-duration contractions that maintain a baseline tone and regulate the pressure within hollow organs and sphincters.

- **Answer:**

11. Norepinephrine released by sympathetic nerves can inhibit gut motility both by directly hyperpolarizing smooth muscle cells and by blocking acetylcholine release from enteric neurons.

- **Answer:**

12. Motilin is an enterogastrone hormone released during a heavy meal to inhibit the migrating motor complex, allowing food to remain in the small intestine for proper mixing.

- **Answer:**

13. In a patient with Hirschsprung's disease, the aganglionic segment of the colon is permanently relaxed and dilated, allowing fecal matter to pass through with zero resistance.

- **Answer:**

14. Cholecystokinin (CCK) slows down digestion in the stomach by contracting the pyloric sphincter and delaying gastric emptying, ensuring adequate time for fat emulsification in the duodenum.

- **Answer:**

15. Vasodilation of the splanchnic circulation during digestion is driven synergistically by metabolic demand, a drop in local oxygen levels, and the release of gastrointestinal hormones like secretin and CCK.

- **Answer:**